

21 C Bulb X : $R_x = 10^2 / 20 = 5.0 \, \Omega$

Bulb Y : $R_y = 5^2 / 2 = 12.5 \, \Omega$

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p.d. across X = $5.0 / (12.5 + 5.0) \times 15.0 \text{ V} = 4.3 \text{ V}$

power dissipated in X < 20 W

p.d. across Y = $12.0 / (12.5 + 5.0) \times 15.0 \text{ V} = 10.7 \text{ V}$

power dissipated in Y > 2 W